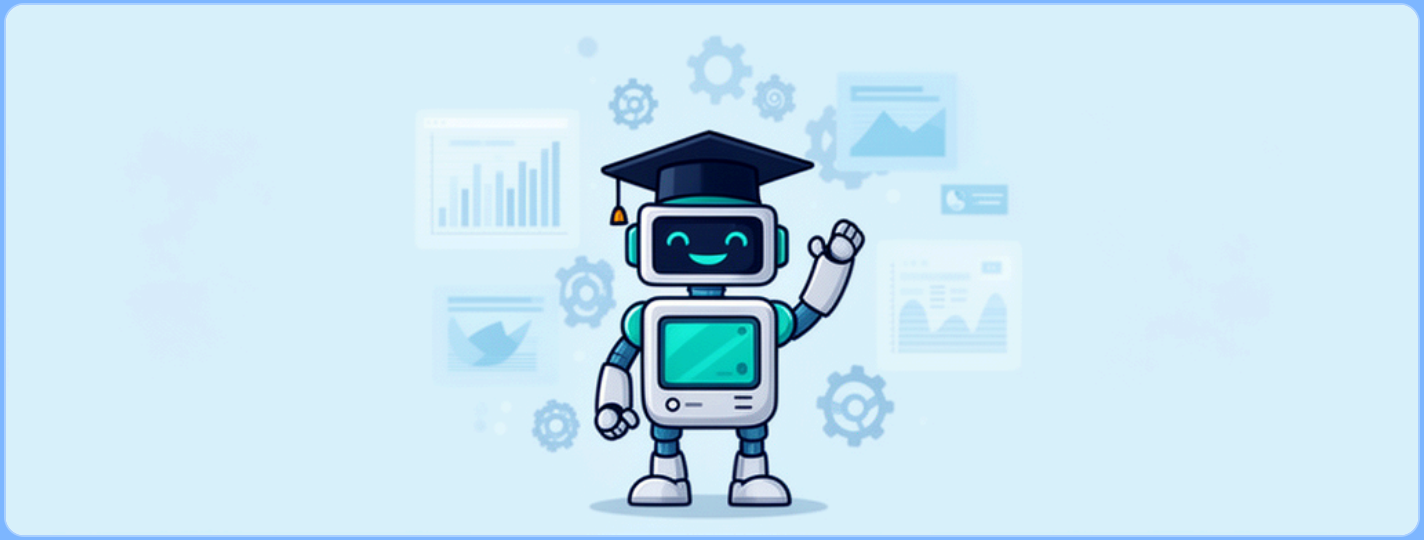
?What is Machine Learning

Day One — You Build an AI

Block 1 · Session 1

.By the end of today, YOU will train a working ML model



The AI Special

.One demo. One red line. One question

.Watch closely

The Animal Game

.The machine sees only this: **8 labeled animals**

Each animal = two numbers (weight, ear length) + one answer

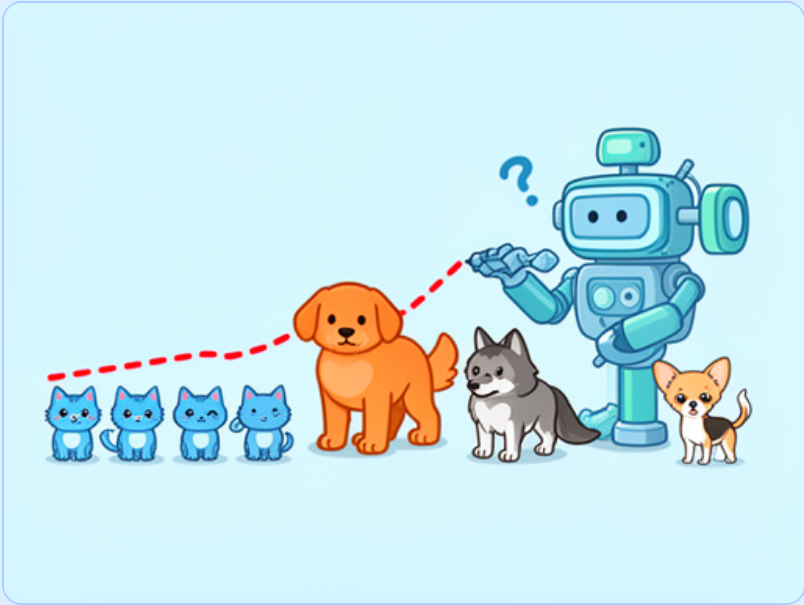
.No rules were given to it. None

The Machine's Line

.From those 8 rows, the machine drew a line **all by itself**

.Behind the line: cats. In front: dogs

"!?Who drew this line"



Now Let's Test It

:Animals the machine has **never seen**

!Wolf ← dog. Neighbor's cat ← cat. **Correct**

!?Puppy ← cat?! Elephant ← dog

"?Its whole secret: one question — "weight ≤ 14 kg

The Big Question

?How do we make computers smart

Way 1 — The Old Way: RULES

`data + RULES → answers`

.Humans write every rule by hand

"IF whiskers AND purrs THEN cat"

The Problem with Rules

.You try: write rules to recognize a cat

!?Counterexamples welcome: hairless cats? cats in costumes? corgis

.Humans can't write enough rules



Way 2 — Machine Learning

`data + answers → RULES`

.We show examples. The machine finds the rules

.That red line at the start? This

AI vs ML

AI = the dream (machines acting smart)

ML = the main way we get there (learning from data)

The Fruit Game

Fruit ←	Texture	Size	Color
?	smooth	medium	red
?	smooth	long	yellow
?	wrinkly	small	purple

.Cover the answer column. Guess the fruit from the clues alone

New Vocabulary

Feature = a clue *(a column)*

Label = the answer *(the column we predict)*

Example = one row = clues + answer

?Which Feature Matters

?Color? Size? Texture

?Which clue is most useful for guessing the fruit

(.Vote. Debate. There is no wrong answer — yet)

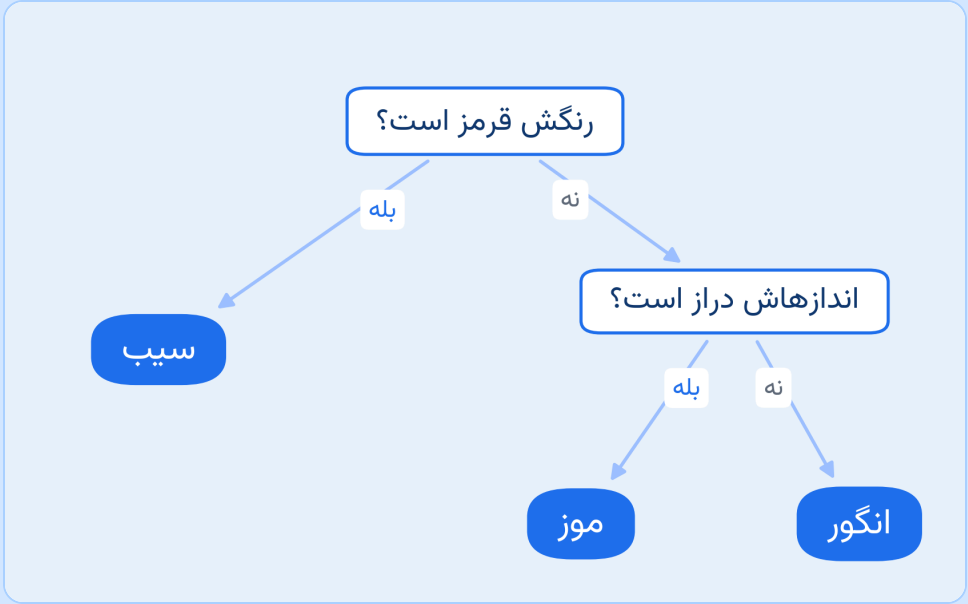
Questions in a Chain

:Line your guesses up into a chain

.ask color first, then size

.A decision map — by hand, no code anywhere

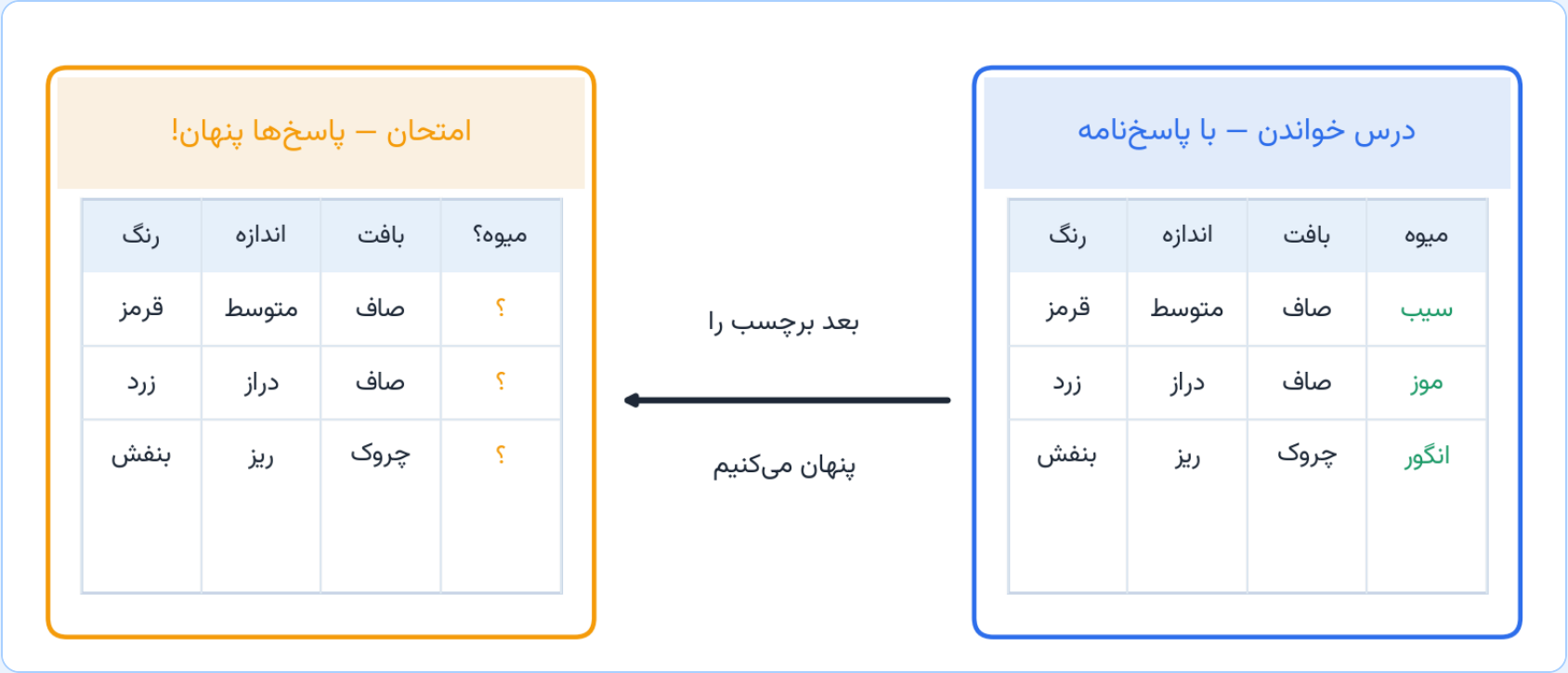
The Fruit Decision Map



.Your own guesses — just tidied up

Supervised Learning

Learning from examples **that include the answers**.



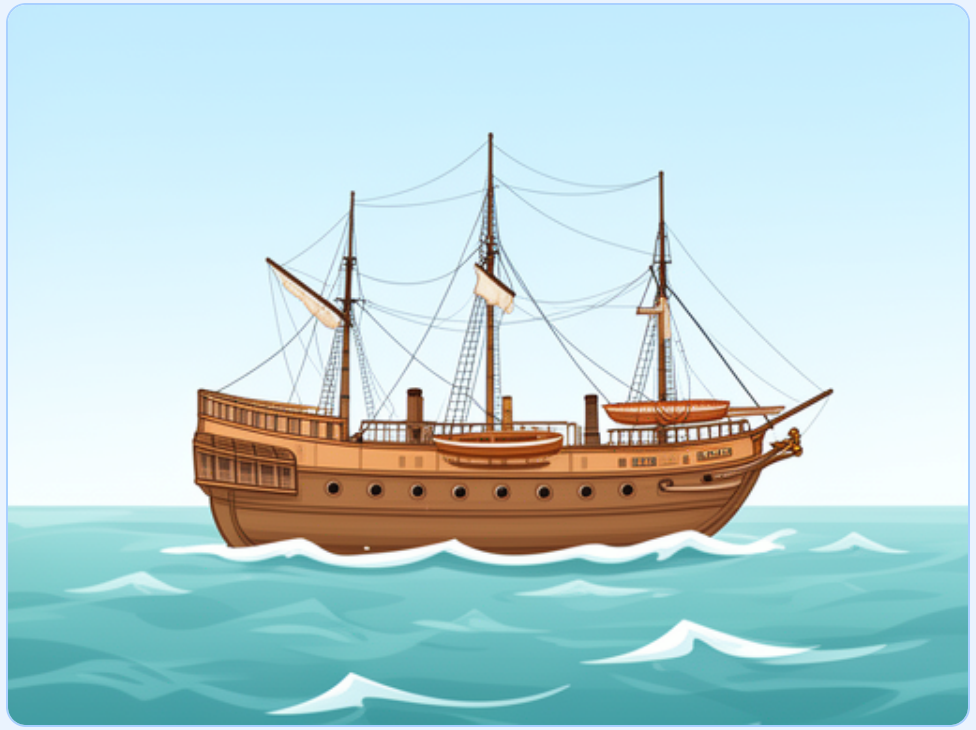
?"Why "Supervised

!No human is watching the model

.It means the **data contains the answers**

(.History picked a weird word)

The Titanic

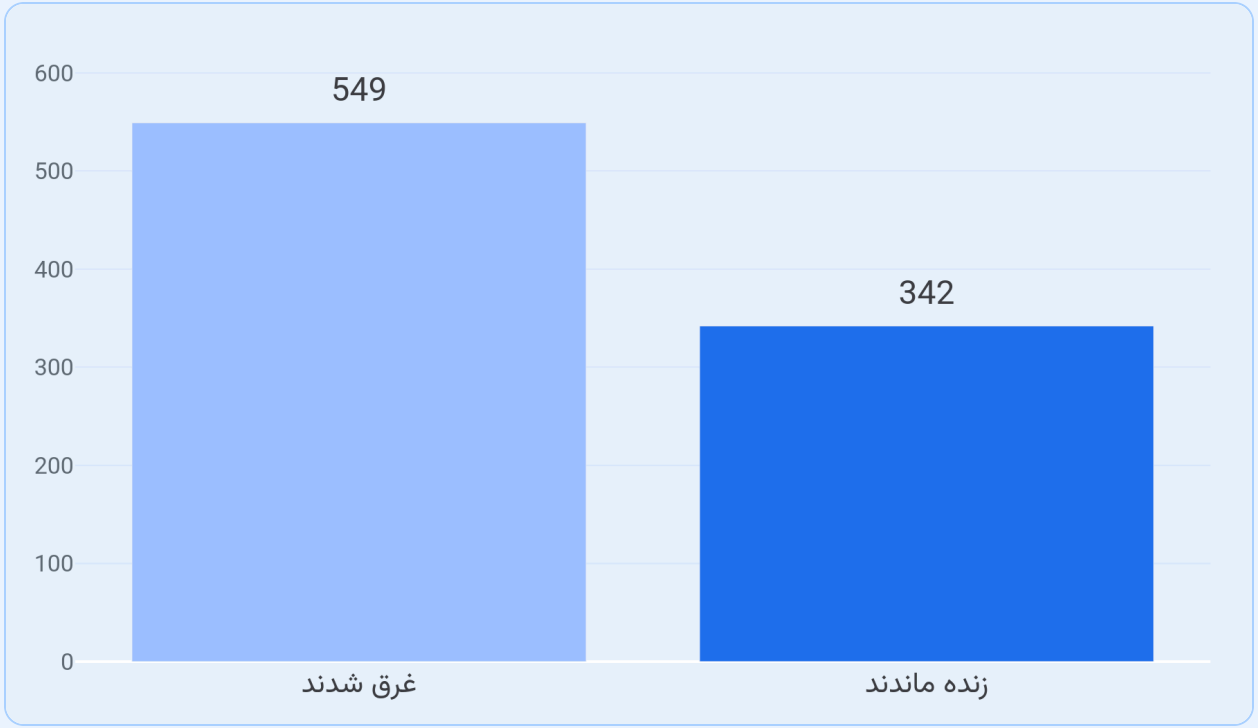


.passengers. 891 rows 891

.Every row is a life. Every column is a clue

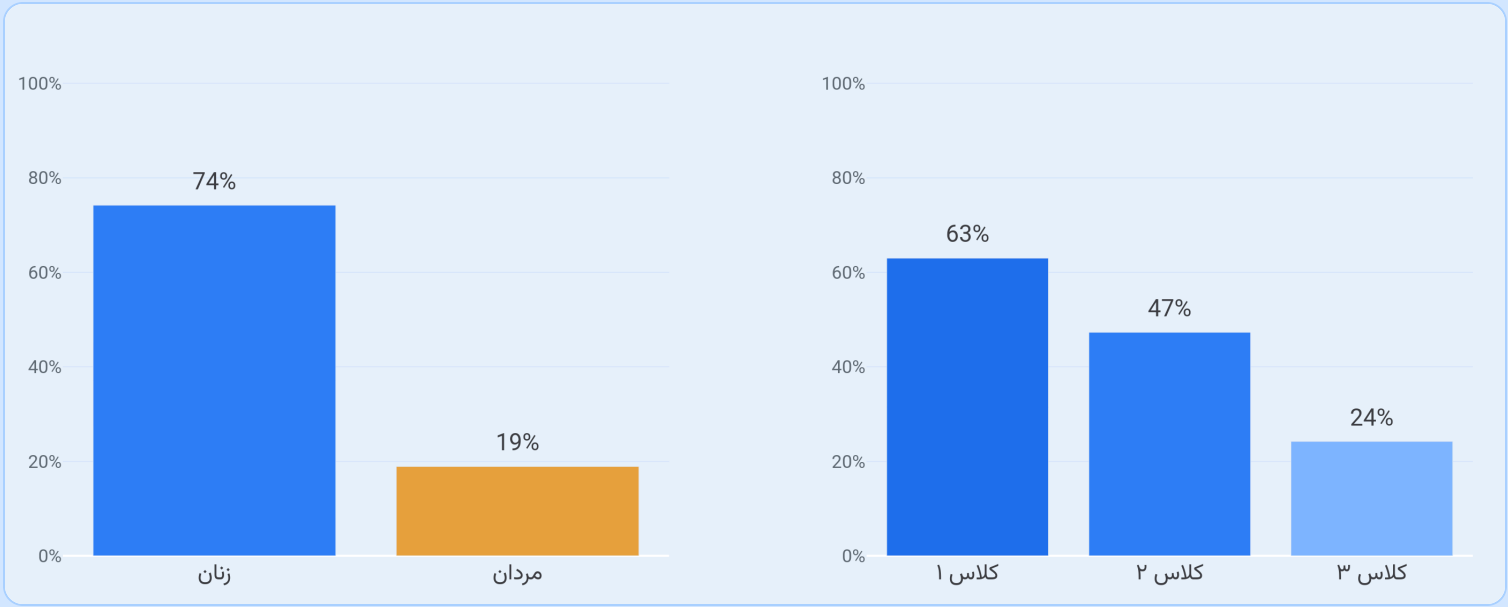
?Would YOU have survived

Data as Pictures



.Always look at your data before doing anything with it

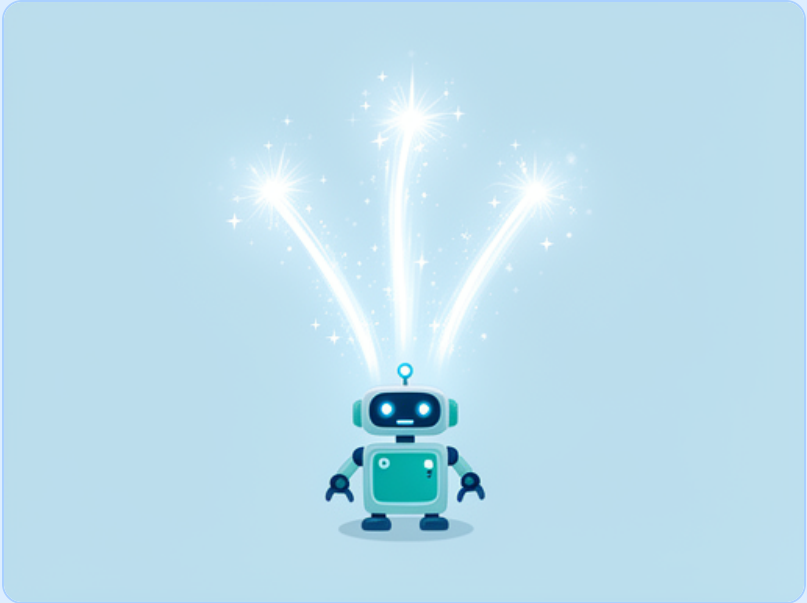
Survival by Sex & Class



.of women lived. Only ~20% of men did 75%~

THE MOMENT

Train an AI in 3 lines



.Type with me. Really — open the notebook and type

The 3 Magic Lines

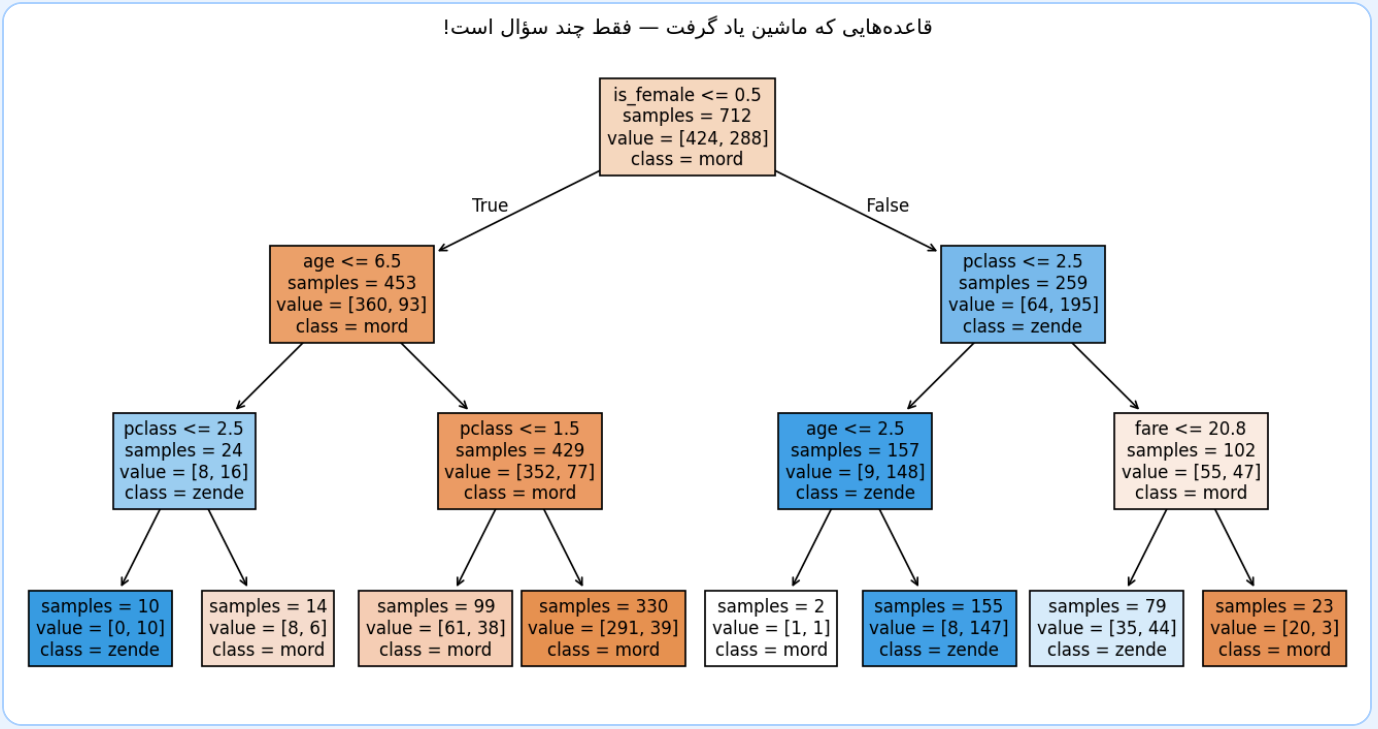
~/ml-course

```
model = DecisionTreeClassifier(max_depth=3) # 1. pick a student ← this IS the model
model.fit(X, y) # 2. let it study
model.predict(new_passengers) # 3. ask it questions
```

create → fit → predict

.This rhythm is ALL of ML. Not just today

What It Learned



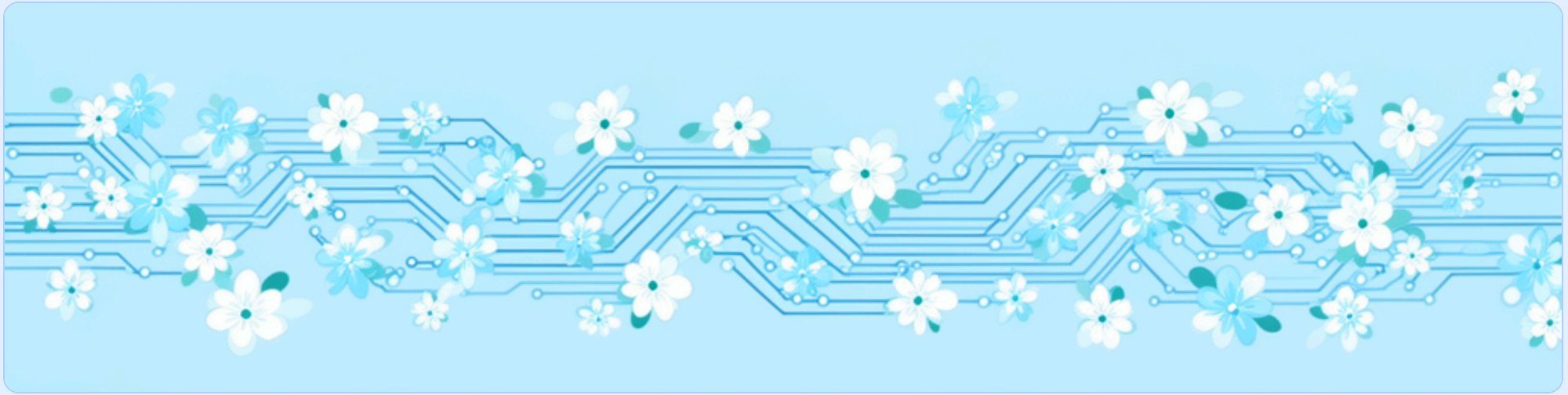
.It's just questions. You could draw this

One Last Thing

.The model is **not a brain**

.It found question-patterns matching survival

.That's already amazing enough



Today You Learned

- Rules → answers, or answers → **rules**
- **Features** (clues) & **labels** (answers)
- **Supervised** learning = examples with answers
- magic lines: **create → fit → predict** 3

Your Ship Tonight

Your first ML notebook → **your GitHub repo**

Would I have survived the Titanic?" — an app you built**"**

At-Home Problems (4)

B1 The Feature Detective **.1**

B2 Build Your Own Dataset **.2**

B3 The 3 Magic Lines on YOUR animals **.3**

B4 The Survival Predictor **.4**

instant score, anytime ← `python judge/run.py s01_notebook.ipynb`

Next Time

.We solve YOUR problems — live

.Bring your attempts. Bring your bugs

.They are the fuel of the next session

